# Georgios Petrakis

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**Profile:** Informatics and Telecommunications graduate focused on the security of networked and distributed systems. Creator of Efaeto; research experience in digital identity and privacy-preserving access control.

## Education

**University of Ioannina** — Arta, Greece
*BSc in Informatics and Telecommunications* — *Sep 2022 to Jun 2026*

- **Final grade: 7.11/10 (Very Good).** 240 ECTS required; 250 recorded. Completed 26 Jun 2026; awarded 2 Oct 2026.
- **Thesis (Parts I and II: 10/10):** "Design and Implementation of a High-Throughput Order Management Application for Large-Scale Events".
- **Coursework:** IoT, Sensor Networks, Special Topics in Networks and Advanced Topics in Network Programming (10/10); Network Management and Statistical Machine Learning (9/10). Also Information System Security, Computer Networks, Operating Systems.

**Certificate of Pedagogical and Teaching Competence**, University of Ioannina, issued 2 October 2026. Planned lessons and taught GLOSSA programming to Grade 11 pupils in Arta; **teaching practicum: 9.8/10**.
**Computer and Network Technician, Level 4**, 5th Vocational High School of Heraklion, Sep 2019 to Jun 2022. **Grade: 18.8/20**.

## Experience

**Research Intern, Secure Software Development (full-time)** — Mar to Apr 2026
*Knowledge and Intelligent Computing (KIC) Laboratory, University of Ioannina* — *Arta, Greece*

- Developed the full web application and its security: React frontend, Node.js/Express backend, MongoDB storage and API integration for Roblox authentication, age verification and access control.
- Implemented digital credentials and single-use tokens using the eIDAS issuer/holder/verifier model, with rate limiting and protection against NoSQL injection and XSS.
- Contributed to research writing on security architecture and the protection of minors in virtual environments.

**Efaeto** — Ongoing project
*Creator and Software Developer* — *Heraklion, Greece*

- Grew my thesis system into an ordering platform for restaurants, cafés, beach bars and festivals, which I build, install and support myself. It keeps taking orders without internet and has staff role permissions and encrypted backups.
- Used by the Avgeniki Cultural Association in Heraklion at its 2025 and 2026 events, with several tills and separate kitchen and drinks displays.

## Research and Projects

**Local-First Event Order Management System** | *Bachelor's Thesis*

- Built an event-driven MERN system for order management at large events with role-based access control, Socket.IO synchronisation and ESC/POS printing over raw TCP with Greek character support.
- Deployed for **seven operational days** at a local cultural festival, recording **694 transactions** (over EUR 22,000) without errors.
- JMeter load test on localhost: **178 ms mean response time**, 1.58M requests from **200 simulated clients** over 45 min.
- Security and resilience tests: OWASP ZAP (injection, XSS), up to 80% packet loss, ten-terminal race for the last stock unit, failover with resync in about 310 ms.

**Arduino WSN ESPNow:** Built an ESP32 sensor network in C++ using ESP-NOW, GPIO control, hardware interrupts and state-machine logic.
**Social Network Platform:** Built a React, Node.js, Express and MongoDB application with REST APIs, persistent storage, authentication and authorisation.
**Conference paper (presented):** "A Local-First Event-Driven Order Management System for Temporary Events", **IEEE SEEDA-CECNSM 2026**, Aristotle University of Thessaloniki, 18 to 20 September 2026. Co-authored with Spiridoula V. Margariti and Christos Gogos.
**Research manuscript:** "Bridging the Age Enforcement Gap in Metaverse Platforms: A Privacy-Preserving Age-Aware Access Control Architecture for Child Protection". Co-author; developed the web application and implemented its security controls. Manuscript in preparation for submission to *Computers & Security*.

## Technical Skills

**Programming**: JavaScript, TypeScript, Python, C/C++, SQL, Lua, VHDL, PHP
**Web and data**: React, Node.js, Express, Socket.IO/WebSockets, Tailwind CSS, MongoDB, SQLite
**Tools**: Git, Docker, Linux, Wireshark, Cisco Packet Tracer, Roblox Studio, Arduino IDE

## Languages and Qualifications

**Greek:** Native. **English:** C2, LRN Level 3 Certificate in ESOL International (CEF C2), issued 18 September 2018. **IELTS Academic** booked for 5 December 2026.
**Develop AI-Powered Prototypes in Google AI Studio**, Google Cloud Skills Boost, May 2026.
**Introduction to Cybersecurity; NDG Linux Unhatched**, Cisco Networking Academy, February 2023.
**State Certificate of Informatics**, Greek Ministry of Education, July 2022.